STA2023 Exam 2 — SIDE A — Q1 TABLE · Q2 VALID? · Q3 DISCRETE · Q4 BINOMIAL — every scenario solved

■ GREEN = the question ■ BLUE = the words → what to do ■ BLACK = the math ■ RED = recipe / answer

HOW TO USE: find the question type → read its RECIPE → find the lettered part whose words match your question → copy its steps with your numbers. Probabilities: 4 decimals (0.4 → 0.4000); μ, σ, x: 2 decimals. Show the formula, box the answer.

Q1 — TWO-WAY TABLE (or / and / given / not / two picks)

"The table below summarizes the employment status of 78 residents."

Status	Male	Female	Total
Full-time	22	18	40
Part-time	6	10	16
Unemployed	3	2	5
Not in labor force	5	12	17
Total	**36**	**42**	**78**

RECIPE
1. **Total every row and column** first (if the counts come in words, draw the table).
2. **is A** → A's total ÷ 78 · **A or B** → add both totals, subtract the overlap cell once · **A and B** ("a part-time female") → the one cell ÷ 78 · **A given B** ("of the females", "among the males", "if female") → the cell ÷ B's total, not 78 · **not A** → 1 − P(A).
3. Fraction → decimal (4 places) → box it.

(a) probability a randomly selected resident is male
one category → its total ÷ grand total

$$P(\text{male}) = \frac{36}{78} = 0.4615$$ **0.4615**

(b) …is part-time
a row → add its cells, ÷ 78

$$P(\text{part-time}) = \frac{6+10}{78} = \frac{16}{78} = 0.2051$$ **0.2051**

(c) …is male OR unemployed
OR → add both totals, subtract the overlap (unemployed males = 3) once

$$P(\text{M or U}) = P(\text{M}) + P(\text{U}) - P(\text{M and U})$$
$$= \frac{36}{78} + \frac{5}{78} - \frac{3}{78} = \frac{38}{78} = 0.4872$$ **0.4872**

Watch out: forgetting the overlap gives 41/78 = 0.5256 × (check: 36 males + 2 unemployed women = 38 people ✓)

(d) …is female AND not in the labor force
AND → the one cell where the row and column meet, ÷ 78

$$P(\text{F and NLF}) = \frac{12}{78} = 0.1538$$ **0.1538**

(e) …is part-time GIVEN the resident is female
GIVEN → divide by the given group (the 42 females), not 78

$$P(\text{PT} \mid \text{F}) = \frac{\text{part-time females}}{\text{all females}} = \frac{10}{42} = 0.2381$$ **0.2381**

Watch out: 10/78 = 0.1282 is the classic error — the word after "given" is the denominator.

(f) …is NOT male
NOT → 1 − P(male)

$$P(\text{not male}) = 1 - \frac{36}{78} = \frac{42}{78} = 0.5385$$ **0.5385**

(g) …is full-time OR part-time
two rows of the same variable can't overlap → just add

$$P(\text{FT or PT}) = \frac{40}{78} + \frac{16}{78} = \frac{56}{78} = 0.7179$$ **0.7179**

(h) …is female GIVEN part-time · unemployed GIVEN male
the given group is now the 16 part-timers · the 36 males

$$P(\text{F} \mid \text{PT}) = \frac{10}{16} = 0.6250 \qquad P(\text{U} \mid \text{M}) = \frac{3}{36} = 0.0833$$
0.6250 · 0.0833

(i) …is neither male nor unemployed
NOT (M or U) → 1 − the answer to (c)

$$1 - \frac{38}{78} = \frac{40}{78} = 0.5128$$ **0.5128**

(j) are "part-time" and "female" independent?
independent = knowing one doesn't change the other: compare P(PT | F) with P(PT)

$$P(\text{PT} \mid \text{F}) = \frac{10}{42} = 0.2381 \qquad P(\text{PT}) = \frac{16}{78} = 0.2051$$
$0.2381 \neq 0.2051$ **not independent (dependent)**

equal → independent. Disjoint (mutually exclusive) = no shared cell: full-time & unemployed → disjoint; male & unemployed share the cell 3 → not disjoint.

(k) two residents picked without replacement: both male · WITH replacement: both male
AND for two picks → multiply; without replacement the 2nd pick has one fewer male and one fewer person; with replacement it is the same fraction twice

$$\frac{36}{78} \cdot \frac{35}{77} = 0.2098 \qquad \frac{36}{78} \cdot \frac{36}{78} = 0.2130$$ **0.2098 · 0.2130**

(l) two picks: at least one male · at most one male
at least one → 1 − P(none) = 1 − P(both female) · at most one → 1 − P(both male)

$$1 - \frac{42}{78} \cdot \frac{41}{77} = 1 - 0.2867 = 0.7133 \qquad 1 - 0.2098 = 0.7902$$
0.7133 · 0.7902

(m) two picks: exactly one male
male-then-female OR female-then-male → add the two orders

$$\frac{36}{78} \cdot \frac{42}{77} + \frac{42}{78} \cdot \frac{36}{77} = \frac{1512}{6006} + \frac{1512}{6006}$$
$$= \frac{3024}{6006} = 0.5035$$ **0.5035**

Q2 — VALID PROBABILITY DISTRIBUTION?

"Determine whether the following distribution represents a valid discrete probability distribution. If not, explain why." x: 0, 1, 2, 3, 4 with P(x): 0.34, 0.21, 0.13, 0.04, 0.01

RECIPE
1. **Rule 1:** every P(x) between 0 and 1 (0 and 1 allowed). **Rule 2:** the P(x) add to exactly 1 (fractions or percents → decimals first; 0.9999 from rounding counts).
2. Write BOTH checks and a sentence: valid, or NOT valid and which rule fails.

(a) the exam's table
rule 1: 0.34, 0.21, 0.13, 0.04, 0.01 are all between 0 and 1 ✓ · rule 2: add them

$$0.34 + 0.21 + 0.13 + 0.04 + 0.01 = 0.73 \neq 1$$
NOT valid — adds to 0.73, not 1

(b) if the sum is more than 1 (e.g. 0.15, 0.25, 0.30, 0.20, 0.12)
rule 1 ✓ · rule 2:

$$0.15 + 0.25 + 0.30 + 0.20 + 0.12 = 1.02 \neq 1$$
NOT valid — adds to 1.02

(c) if one entry is negative (e.g. 0.5, 0.4, 0.2, −0.1)
rule 1 fails: −0.1 < 0, even though the sum is 1

$$0.5 + 0.4 + 0.2 + (-0.1) = 1 \text{ but the last entry } -0.1 < 0$$
NOT valid — a probability can't be negative

(d) if everything checks out (e.g. 0.2, 0.4, 0.3, 0.1)
rule 1 ✓ · rule 2:

$$0.2 + 0.4 + 0.3 + 0.1 = 1$$
valid — each P(x) is 0–1 and they add to 1

(e) find the missing probability (0.34, 0.21, 0.13, 0.04, ?)
the blank must make the total exactly 1

$$P(4) = 1 - (0.34 + 0.21 + 0.13 + 0.04) = 1 - 0.72 = 0.28$$ **0.28**

Q3 — DISCRETE DISTRIBUTION (Stanley Cup games)

"The following data represent the number of games played, X, in the Stanley Cup before a champion was determined." x: 4, 5, 6, 7 with frequency 20, 20, 26, 17.

RECIPE
1. **P(x) = frequency ÷ total.** Mean μ = Σ x·P(x). SD: (x − μ)²·P(x) for each x, add (= variance), √.
2. **Any probability:** words → the list of x. **at least / at most / "or more" / "or less" keep the number IN; more than / fewer than leave it OUT** (P(X ≤ 5) = at most 5, P(X ≥ 6) = at least 6, P(X < 6) = fewer than 6, P(X > 6) = more than 6). Then add their P(x).

(a) construct the probability distribution
total = 20 + 20 + 26 + 17 = 83; each P(x) = its count ÷ 83

$$P(4) = \frac{20}{83} = 0.2410 \qquad P(5) = \frac{20}{83} = 0.2410$$
$$P(6) = \frac{26}{83} = 0.3133 \qquad P(7) = \frac{17}{83} = 0.2048$$

check: 0.2410 + 0.2410 + 0.3133 + 0.2048 = 1.0001 ≈ 1 (rounding) ✓

(b) graph the distribution
x across, P(x) up, a bar above each x with height P(x), label both axes (tallest bar = x = 6)



(c) compute and interpret the mean
each x times its P(x), then add

$$\mu = \sum x \cdot P(x) = 4(0.2410) + 5(0.2410) + 6(0.3133) + 7(0.2048)$$
$$= 0.9640 + 1.2050 + 1.8798 + 1.4336 = 5.48$$ **μ = 5.48 games**

interpret: "over many series, the average length is about 5.48 games."

(d) compute the standard deviation
subtract μ = 5.48, square, × P(x), add, √

$$(4 - 5.48)^2(0.2410) = 2.1904(0.2410) = 0.5279$$
$$(5 - 5.48)^2(0.2410) = 0.2304(0.2410) = 0.0555$$
$$(6 - 5.48)^2(0.3133) = 0.2704(0.3133) = 0.0847$$
$$(7 - 5.48)^2(0.2048) = 2.3104(0.2048) = 0.4732$$
$$\sigma^2 = 0.5279 + 0.0555 + 0.0847 + 0.4732 = 1.1413$$
$$\sigma = \sqrt{1.1413} = 1.07$$ **σ = 1.07 games**

Watch out: if they ask for the variance, stop at 1.1413. No ÷ 4 and no ÷ 83 here.

(e) P(exactly 6) · P(at least 6) = 6 or more
one value → x = 4 5 **6** 7 · 6 is IN → x = 4 5 **6 7**

$$P(6) = 0.3133$$ **0.3133**
$$P(X \geq 6) = 0.3133 + 0.2048 = 0.5181$$ **0.5181**

(f) P(more than 6) · P(fewer than 6) = under 6
6 is OUT → x = 4 5 6 **7** · 6 is OUT → x = **4 5** 6 7

$$P(X > 6) = 0.2048$$ **0.2048**
$$P(X < 6) = 0.2410 + 0.2410 = 0.4820$$ **0.4820**

(g) P(at most 5) = 5 or fewer · P(at most 6) = fewer than 7
5 is IN (same list as fewer than 6) → x = **4 5** 6 7 · 6 is IN → x = **4 5 6** 7

$$P(X \leq 5) = 0.2410 + 0.2410 = 0.4820$$ **0.4820**
$$P(X \leq 6) = 0.2410 + 0.2410 + 0.3133 = 0.7953$$ **0.7953**

(h) P(at least 5) = more than 4 · P(between 5 and 6 inclusive)
5 is IN → x = 4 **5 6 7** · both ends IN → x = 4 **5 6** 7

$$P(X \geq 5) = 0.2410 + 0.3133 + 0.2048 = 0.7591$$ **0.7591**
$$P(5 \leq X \leq 6) = 0.2410 + 0.3133 = 0.5543$$ **0.5543**

(i) P(between 4 and 7, NOT inclusive) · P(5 or 7)
ends OUT → x = 4 **5 6** 7 · two values → x = 4 **5** 6 **7**

$$P(4 < X < 7) = 0.2410 + 0.3133 = 0.5543$$ **0.5543**
$$P(5 \text{ or } 7) = 0.2410 + 0.2048 = 0.4458$$ **0.4458**

(j) P(not 4) · P(not 6)
NOT → 1 − that value

$$1 - P(4) = 1 - 0.2410 = 0.7590$$ **0.7590**
$$1 - P(6) = 1 - 0.3133 = 0.6867$$ **0.6867**

adding the leftovers instead (0.2410 + 0.3133 + 0.2048 = 0.7591) is 0.0001 off only from rounding — both accepted.

(k) most likely number of games · is 7 games unusual? · in 50 series, how many go 7?
mode = the biggest P(x) = 0.3133 → **6 games** · unusual: P(7) = 0.2048 is not below 0.05, and μ ± 2σ = 5.48 ± 2(1.07) = 3.34 to 7.62 holds 7 → **not unusual** · expected count = 50 × 0.2048 = 10.24 ≈ **10**

Q4 — BINOMIAL (60% say yes; sample of 10)

"According to a Gallup poll, 60% of U.S. women 18 years old or older stated that the minimum driving age should be 18. In a random sample of 10 U.S. women 18 years old or older, find the probability that:"

RECIPE
1. **Set up:** n = sample size = 10, p = the percent as a decimal = 0.6, 1 − p = 0.4, X = number who say yes.
2. **Words → list of x:** at most / at least / "or fewer" / "or more" keep the number IN; fewer than / more than leave it OUT; exactly = one value; at least one = 1 − P(0); "3 say NO" = 7 say yes.
3. **Short list (1–4 values) → compute each P(x) and add. Long list → compute the OTHER side and do 1 − (that).**
4. **Each P(x):** $P(x) = {}_nC_x\, p^x (1-p)^{n-x}$, ${}_nC_x = \frac{n!}{x!\,(n-x)!}$ (TI-89: 2nd MATH → Probability → nCr(10,3) = 120). 4 decimals, box it.

(a) exactly 3 believe the minimum driving age should be 18
one value → the formula once

$${}_{10}C_3 = \frac{10!}{3!\,7!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$
$$(0.6)^3 = 0.216 \qquad (0.4)^7 = 0.0016384$$
$$P(3) = {}_{10}C_3(0.6)^3(0.4)^7 = 120(0.216)(0.0016384) = 0.0425$$ **0.0425**

Watch out: the power on 0.4 is n − x = 7, not 3.

(b) fewer than 3 believe…
"fewer than 3" leaves 3 OUT → x = **0 1 2** 3 4 5 6 7 8 9 10 → short list → add the three

$${}_{10}C_0 = \frac{10!}{0!\,10!} = 1 \qquad {}_{10}C_1 = \frac{10!}{1!\,9!} = 10$$
$${}_{10}C_2 = \frac{10!}{2!\,8!} = \frac{10 \cdot 9}{2 \cdot 1} = 45$$
$$P(0) = {}_{10}C_0(0.6)^0(0.4)^{10} = 1(1)(0.0001048576) = 0.0001$$
$$P(1) = {}_{10}C_1(0.6)^1(0.4)^9 = 10(0.6)(0.000262144) = 0.0016$$
$$P(2) = {}_{10}C_2(0.6)^2(0.4)^8 = 45(0.36)(0.00065536) = 0.0106$$
$$P(X < 3) = 0.0001 + 0.0016 + 0.0106 = 0.0123$$ **0.0123**

(c) between 1 and 3 (inclusive) believe…
both ends IN → x = 0 **1 2 3** 4 5 6 7 8 9 10 → add P(1), P(2) from (b) and P(3) from (a)

$$P(1 \leq X \leq 3) = 0.0016 + 0.0106 + 0.0425 = 0.0547$$ **0.0547**

(d) sample of 200: expected number and standard deviation
binomial shortcuts (no table): μ = np, σ = √(np(1 − p))

$$\mu = np = 200(0.6) = 120$$
$$\sigma = \sqrt{np(1-p)} = \sqrt{200(0.6)(0.4)} = \sqrt{48} = 6.93$$ **μ = 120, σ = 6.93**

Watch out: 48 is the variance — take the √. For the 10: μ = 10(0.6) = 6, σ = √2.4 = 1.55.

(e) at most 3 = 3 or fewer = no more than 3
3 is IN → x = **0 1 2 3** 4 5 6 7 8 9 10 → add (b)'s three and (a)

$$P(X \leq 3) = 0.0001 + 0.0016 + 0.0106 + 0.0425 = 0.0548$$ **0.0548**

(f) more than 3 = over 3
3 is OUT → x = 0 1 2 3 **4 5 6 7 8 9 10** → long list → 1 − the other side (x = 0, 1, 2, 3 = part e)

$$P(X > 3) = 1 - P(X \leq 3) = 1 - 0.0548 = 0.9452$$ **0.9452**

(g) at least 3 = 3 or more
3 is IN → x = 0 1 2 **3 4 5 6 7 8 9 10** → long list → 1 − the other side (x = 0, 1, 2 = part b)

$$P(X \geq 3) = 1 - P(X \leq 2) = 1 - 0.0123 = 0.9877$$ **0.9877**

Watch out: "more than 3" subtracts 0.0548 (3 goes to the other side); "at least 3" subtracts 0.0123.

(h) between 1 and 3 NOT inclusive
ends OUT → only x = 2

$$P(2) = 0.0106$$ **0.0106**

(i) at least one · none · all 10
at least one → 1 − P(0) · none → P(0) · all → P(10) = $(0.6)^{10}$

$$P(X \geq 1) = 1 - P(0) = 1 - 0.0001 = 0.9999$$
$$P(0) = (0.4)^{10} = 0.0001 \qquad P(10) = (0.6)^{10} = 0.0060$$
0.9999 · 0.0001 · 0.0060

(j) at most 8 · more than 8
at most 8: 8 is IN → x = 0 … 8 is long → 1 − P(9) − P(10). more than 8: x = 9, 10 → add

$$P(9) = {}_{10}C_9(0.6)^9(0.4)^1 = 10(0.010077696)(0.4) = 0.0403$$
$$P(10) = {}_{10}C_{10}(0.6)^{10}(0.4)^0 = 1(0.0060466176)(1) = 0.0060$$
$$P(X > 8) = 0.0403 + 0.0060 = 0.0463$$
$$P(X \leq 8) = 1 - 0.0463 = 0.9537$$ **0.0463 · 0.9537**

(k) exactly 6 · at least 8 (middle values)
the other values, same formula: P(4) = 0.1115, P(5) = 0.2007, P(6) = 0.2508, P(7) = 0.2150, P(8) = 0.1209

$$P(6) = 0.2508 \qquad P(X \geq 8) = 0.1209 + 0.0403 + 0.0060 = 0.1672$$
0.2508 · 0.1672

(l) not all say yes · exactly 3 say NO
NOT → 1 − P(10) · 3 "no" = 7 "yes" → P(7) (same as switching to p = 0.4)

$$P(\text{not all}) = 1 - P(10) = 1 - 0.0060 = 0.9940$$
$${}_{10}C_3(0.4)^3(0.6)^7 = 120(0.064)(0.0279936) = 0.2150$$ **0.9940 · 0.2150**

(m) is 130 of 200 unusual? · would only 2 of 10 be unusual?
count: usual range μ ± 2σ (needs np(1 − p) = 48 ≥ 10 ✓) · probability: "2 or fewer" below 0.05?

$$\mu \pm 2\sigma = 120 \pm 2(6.93) = 106.1 \text{ to } 133.9 \text{ (130 is inside)}$$
$$P(X \leq 2) = 0.0001 + 0.0016 + 0.0106 = 0.0123 < 0.05$$
130 not unusual · 2 of 10 unusual

(100 of 200 would be below 106.1 → unusual; 9 of 10: P(X ≥ 9) = 0.0463 < 0.05 → unusual)

(n) explain why this is a binomial experiment
1) fixed n = 10 trials · 2) each trial yes/no · 3) same p = 0.6 each trial · 4) independent (random sample). Not binomial: n not fixed ("until…") or no-replacement picks over 5% of the group.

STA2023 Exam 2 — SIDE B — Q5 UNIFORM · Q6 z · Q7 NORMAL — every scenario solved

■ GREEN = the question ■ BLUE = the words → what to do ■ BLACK = the math ■ RED = recipe / answer

Q5 — UNIFORM (temperature 70°F to 100°F)

"The daytime temperature X (°F) in a coastal city during summer is uniformly distributed between 70°F and 100°F."

RECIPE

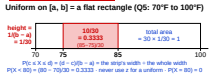
1. **Set up:** a = smaller end 70, b = bigger end 100, width b − a = 30, height 1/(b − a) = 1/30 (area = 30 × 1/30 = 1).
2. **Words → the piece from c to d:** "below 80" = 70 to 80 · "above 90" = 90 to 100 · outside 70–100 is cut off at 70 or 100 · one exact value → 0. **P = (d − c) ÷ (b − a)**, 4 decimals.
3. **Backwards** (a percent is given): x = a + (area to the LEFT) (b − a). No z, no hints in this question.

(a) sketch / describe the pdf

a flat line from 70 to 100 at height 1/30, zero elsewhere

$$\text{height} = \frac{1}{b-a} = \frac{1}{100-70} = \frac{1}{30} = 0.0333 \qquad \text{area} = 30 \times \frac{1}{30} = 1$$

height 1/30



(b) P(75 ≤ X ≤ 85)

a middle piece → piece from 75 to 85

$$P(75 \le X \le 85) = \frac{85-75}{100-70} = \frac{10}{30} = 0.3333$$ **0.3333**

(c) P(X < 80) = below 80 = at most 80

the left end up to 80 → piece from 70 to 80

$$P(70 \le X \le 80) = \frac{80-70}{100-70} = \frac{10}{30} = 0.3333$$ **0.3333**

(d) P(X > 90) = above 90 = at least 90

90 up to the right end → piece from 90 to 100

$$P(90 \le X \le 100) = \frac{100-90}{100-70} = \frac{10}{30} = 0.3333$$ **0.3333**

(e) P(X ≥ 95)

95 to 100 → piece from 95 to 100

$$P(95 \le X \le 100) = \frac{100-95}{100-70} = \frac{5}{30} = 0.1667$$ **0.1667**

(f) P(60 ≤ X ≤ 75) — 60 is outside the range

only 70 to 75 exists → piece from 70 to 75

$$P(70 \le X \le 75) = \frac{75-70}{100-70} = \frac{5}{30} = 0.1667$$ **0.1667**

(g) P(X = 80) exactly · P(X < 60) · P(50 < X < 110)

one point has no width → 0 · entirely outside → 0 · the whole range → 1

$$P(X = 80) = \frac{80-80}{30} = 0$$

$$P(X < 60) = 0 \qquad P(50 < X < 110) = 1$$ **0 · 0 · 1**

(h) NOT between 75 and 85 · below 75 OR above 95

NOT → 1 − (b) · OR of two pieces → add them

$$1 - \frac{10}{30} = 0.6667$$

$$\frac{75-70}{30} + \frac{100-95}{30} = \frac{5}{30} + \frac{5}{30} = 0.3333$$ **0.6667 · 0.3333**

(i) P(X > 90 GIVEN X > 80)

GIVEN → divide by the given piece

$$\frac{P(X > 90)}{P(X > 80)} = \frac{10/30}{20/30} = \frac{10}{20} = 0.5000$$ **0.5000**

(j) 20% of days are cooler than what temperature?

backwards: area to the LEFT = 0.20 → x = a + (area)(b − a)

$$x = 70 + 0.20(30) = 70 + 6 = 76 \qquad \text{check: } \frac{76-70}{30} = 0.20 \checkmark$$

76°F

(k) the hottest 10% start at what temperature?

top 10% → 90% below it → left area 0.90

$$x = 70 + 0.90(30) = 70 + 27 = 97 \qquad \text{check: } \frac{100-97}{30} = 0.10 \checkmark$$

97°F

(l) median, quartiles, mean, SD

any percentile: x = 70 + (percent)(30)

$$\text{median} = 70 + 0.5(30) = 85 \qquad Q_1 = 70 + 0.25(30) = 77.5$$

$$Q_3 = 70 + 0.75(30) = 92.5$$

$$\mu = \frac{a+b}{2} = 85 \qquad \sigma = \frac{b-a}{\sqrt{12}} = \frac{30}{\sqrt{12}} = 8.66$$

85 · 77.5 · 92.5 · 8.66

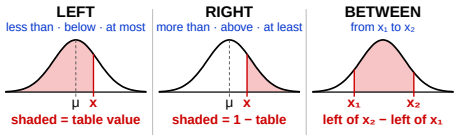
Q6 — STANDARD NORMAL z (the exam's hint = the table value = area to the LEFT)

"Use the standard normal table (on the real exam: the printed z hints). Round to four decimal places." Hints: left of −1.10 = 0.1357 · left of −0.60 = 0.2743 · left of 0.60 = 0.7257 · left of 1.45 = 0.9265 · left of −1.00 = 0.1587 · left of 1.50 = 0.9332

RECIPE

1. **Words → which area:** LEFT of z (below, less than, at most) · RIGHT of z (above, more than, at least, exceeds) · BETWEEN (within) · OUTSIDE (two tails).
2. **Read the hint:** ALWAYS the area to the LEFT of that z, even for a negative z. **LEFT → the hint · RIGHT → 1 − hint · BETWEEN → (left of the bigger z) − (left of the smaller z)**, bigger = further right (−0.60 is bigger than −1.10) · **OUTSIDE → left tail + right tail.**
3. **μ, σ given:** first $z = \frac{x-\mu}{\sigma}$ to 2 decimals for each x (variance given → σ = √variance), then the same rules. **Backwards** (area given): make it a LEFT area → the closest hint/table entry → its z.

READING A PRINTED HINT: "P(Z < 1.67) = 0.9525" or "area to the left of 1.67 is 0.9525" → a LEFT area (below → write it; above → 1 − it) · "P(Z > 1.67) = 0.0475" → already the RIGHT tail (below = 1 − 0.0475) · "P(Z < −1.43) = 0.0764" → already a left area, use as is · "z = 2.05 for the 98th percentile" / "z = 1.28 for the top 10%" → put it in x = μ + zσ (bottom 10% → z = −1.28). Your z not among the hints? Recompute it — a hint exists for every z the exam needs.



(a) area to the left of z = −1.10

LEFT → the hint is the answer (a hint for a negative z is already the left area)

$$P(Z < -1.10) = 0.1357$$ **0.1357**

(b) area between z = −0.60 and z = 1.45

BETWEEN → bigger z is 1.45 (left 0.9265), smaller is −0.60 (left 0.2743)

$$P(-0.60 < Z < 1.45) = 0.9265 - 0.2743 = 0.6522$$ **0.6522**

Watch out: subtract the AREAS, never the z's; a negative answer means you subtracted the wrong way.

(c) μ = 80, σ = 10: P(70 < X < 95)

convert each end to z, then BETWEEN

$$z = \frac{70-80}{10} = -1.00 \to 0.1587 \qquad z = \frac{95-80}{10} = 1.50 \to 0.9332$$

$$P(70 < X < 95) = 0.9332 - 0.1587 = 0.7745$$ **0.7745**

(d) right of −1.10 · right of 1.45 · left of 1.45

RIGHT → 1 − hint · LEFT of a positive z → still just the hint

$$P(Z > -1.10) = 1 - 0.1357 = 0.8643$$

$$P(Z > 1.45) = 1 - 0.9265 = 0.0735$$

$$P(Z < 1.45) = 0.9265$$ **0.8643 · 0.0735 · 0.9265**

(e) between −1.10 and −0.60 (both negative) · between 0.60 and 1.45 (both positive)

same rule: bigger z's left area minus smaller z's

$$0.2743 - 0.1357 = 0.1386 \qquad 0.9265 - 0.7257 = 0.2008$$

0.1386 · 0.2008

(f) outside −1.10 and 1.45 (less than −1.10 OR more than 1.45)

two tails → left tail + right tail

$$0.1357 + (1 - 0.9265) = 0.1357 + 0.0735 = 0.2092$$ **0.2092**

(g) μ = 80, σ = 10: P(X > 95) · P(X ≤ 70)

z = 1.50 → 0.9332 → RIGHT · z = −1.00 → LEFT → the hint

$$P(X > 95) = 1 - 0.9332 = 0.0668 \qquad P(X \le 70) = 0.1587$$

0.0668 · 0.1587

(h) find z with area 0.10 to its LEFT · with 0.025 to its RIGHT

backwards: LEFT area given → closest entry (0.1003) → z · RIGHT given → left = 1 − 0.025 = 0.975 → z

$$z = -1.28 \qquad z = 1.96$$ **−1.28 · 1.96**

(i) the two z's that cut off the middle 90%

0.05 in each tail → left areas 0.05 and 0.95 (0.95 is halfway between 1.64 and 1.65 → 1.645)

$$z = \pm 1.645$$ **±1.645**

(j) μ = 80, σ = 10: bottom 10% cutoff · top 5% · middle 90%

backwards for x: left area → z → x = μ + zσ

$$80 + (-1.28)(10) = 67.2 \qquad 80 + 1.645(10) = 96.45$$

$$80 \pm 1.645(10) = 63.55 \text{ to } 96.45$$ **67.2 · 96.45 · 63.55–96.45**

(k) symmetry · within 1 SD of the mean

P(Z > 1.10) = P(Z < −1.10) = 0.1357 · P(Z < 0) = 0.5000 · left of −z = 1 − (left of z)

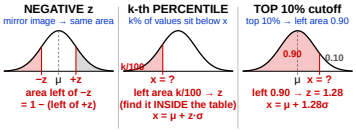
$$P(-1 < Z < 1) = 0.8413 - 0.1587 = 0.6826$$ **0.6826 (the 68%)**

Q7 — NORMAL IN CONTEXT (IQ: μ = 100, σ = 15)

"The Wechsler Intelligence Scale for Children is approximately normally distributed with mean μ = 100 and standard deviation σ = 15."

RECIPE

1. **Forward (an x is given):** $z = \frac{x-\mu}{\sigma}$ to 2 decimals → the hint = area to the LEFT → below: the hint · above: 1 − hint · between: bigger left − smaller left · outside: both tails added.
2. **Backward (a percent / percentile is given):** turn it into the area to the LEFT ("top 10%" → 0.90, "bottom 5%" → 0.05, "k-th percentile" → k/100, "middle 95%" → 0.025 and 0.975) → the hint's z → $x = \mu + z\sigma$ (1 decimal).
3. **Unusual?** tail area < 0.05 or |z| > 2. **Sentence:** "__% of children score …" / "a random child has a 0.__ chance of …".



(a) probability a test taker scores above 125

ABOVE → 1 − (left of z)

$$z = \frac{125-100}{15} = 1.67 \;\to\; \text{hint: left of } 1.67 = 0.9525$$

$$P(X > 125) = 1 - 0.9525 = 0.0475$$ **0.0475**

Watch out: 0.9525 is the #1 wrong answer — fewer than half score above 125, so the answer must be below 0.5.

(b) probability a test taker scores below 90

BELOW → the hint itself

$$z = \frac{90-100}{15} = -0.67 \;\to\; P(X < 90) = 0.2514$$ **0.2514**

(c) proportion scoring between 110 and 140

BETWEEN → two z's, bigger left − smaller left

$$z = \frac{110-100}{15} = 0.67 \to 0.7486 \qquad z = \frac{140-100}{15} = 2.67 \to 0.9962$$

$$P(110 < X < 140) = 0.9962 - 0.7486 = 0.2476$$ **0.2476**

(d) score at the 98th percentile

98% below it → area to the LEFT = 0.98 → hint z = 2.05 → x = μ + zσ

$$x = 100 + 2.05(15) = 100 + 30.75 = 130.75$$ **130.75 ≈ 131**

Watch out: never put 0.98 into the z formula — it is an area, not a score.

(e) outside 70 to 130 (below 70 OR above 130)

two tails → add them

$$z = \frac{70-100}{15} = -2.00 \to 0.0228 \qquad z = \frac{130-100}{15} = 2.00 \to 0.9772$$

$$0.0228 + (1 - 0.9772) = 0.0456$$ **0.0456**

(f) exactly 100 · at most 100 · above 100

one value has no area → 0 · z = 0 → left of 0 = 0.5000 either way

$$P(X = 100) = 0 \qquad P(X \le 100) = P(X > 100) = 0.5000$$

0 · 0.5000

(g) the top 5% cutoff (minimum score for the top 5%)

95% below → left area 0.95 → z = 1.645

$$x = 100 + 1.645(15) = 124.675 \approx 124.7$$ **124.7**

(h) the bottom 10% cutoff (10th percentile)

left area 0.10 → z = −1.28 (keep the minus)

$$x = 100 + (-1.28)(15) = 100 - 19.2 = 80.8$$ **80.8**

(i) the two scores holding the middle 95% · quartiles (middle 50%)

2.5% in each tail → left areas 0.025, 0.975 → z = ±1.96 · quartiles: left 0.25, 0.75 → z = ±0.67

$$100 \pm 1.96(15) = 100 \pm 29.4 \to 70.6 \text{ and } 129.4$$

$$100 \pm 0.67(15) = 100 \pm 10.05 \to 89.95 \text{ and } 110.05$$

70.6–129.4 · 89.95–110.05

(j) the score with 30% above it

30% above → 70% below → left area 0.70 → z = 0.52

$$x = 100 + 0.52(15) = 100 + 7.8 = 107.8$$ **107.8**

(k) a score of 115 is what percentile? · between 85 and 115

forward then read the left area · within 1 SD: z = ±1.00

$$z = \frac{115-100}{15} = 1.00 \to \text{left } 0.8413 \to 84\text{th percentile}$$

$$P(85 < X < 115) = 0.8413 - 0.1587 = 0.6826$$ **84th · 0.6826**

(l) is a score of 65 unusual? · is 140?

tail area below 0.05 (or |z| > 2) → unusual

$$z = \frac{65-100}{15} = -2.33 \to \text{left } 0.0099 < 0.05$$

$$z = \frac{140-100}{15} = 2.67 \to 1 - 0.9962 = 0.0038 < 0.05$$

both unusual

(m) NOT between 110 and 140 · above 125 GIVEN above 100 · 3 children, at least one above 125

NOT → 1 − (c) · GIVEN → divide · at least one → 1 − P(none)

$$1 - 0.2476 = 0.7524 \qquad \frac{0.0475}{0.5000} = 0.0950$$

$$1 - (0.9525)^3 = 1 - 0.8642 = 0.1358$$ **0.7524 · 0.0950 · 0.1358**

(n) interpret "the area above 125 is 0.0475" · a z-score

(1) 4.75% of all children score above 125; (2) a randomly selected child has a 0.0475 chance of scoring above 125 · "z = −2.33" = 2.33 SDs below the mean · "131 is the 98th percentile" = 98% score below 131 · the score 1.5 SDs above the mean = 100 + 1.5(15) = 122.5

Percentile → z: 1st −2.33 · 5th −1.645 · 10th −1.28 · 20th −0.84 · 25th −0.67 · 75th 0.67 · 80th 0.84 · 90th 1.28 · 95th 1.645 · 97.5th 1.96 · 98th 2.05 · 99th 2.33 · middle 80 / 90 / 95 / 99% ↔ ±1.28 / ±1.645 / ±1.96 / ±2.575

CONCEPTS THEY CAN ASK IN WORDS

- **Probability** = long-run fraction of times something happens; 0 ≤ P ≤ 1; **unusual** = below 0.05. **Classical** (equally likely: dice, cards, coins) P = N(E)/N(S), e.g. P(two heads) = 1/4. **Empirical** (from data) P ≈ frequency ÷ trials. **Subjective** = educated guess. **Disjoint** = can't both happen (P(E and F) = 0). **Independent** = one doesn't change the other's probability. Disjoint is NOT independent.
- **Rules:** $P(E^c) = 1 - P(E)$ · at least one = 1 − P(none), e.g. $1 - 0.9986^{1000} = 0.7536$ · $P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$ · $P(E \text{ and } F) = P(E)\,P(F \mid E) \cdot P(F \mid E) = \frac{N(E \text{ and } F)}{N(E)}$. **Counting:** choices in sequence → multiply (26·26·26 = 17,576) · permutation (order matters) ${}_nP_r = \frac{n!}{(n-r)!}$, ${}_7P_5 = 2{,}520$ · combination (order doesn't) ${}_nC_r = \frac{n!}{r!(n-r)!}$, ${}_4C_2 = 6 \cdot 0! = 1$.
- **Discrete** = counted (0, 1, 2, …); **continuous** = measured (time, weight) → probability = area, P(one exact value) = 0. E(X) = Σ x·P(x) = long-run average; insurance: premium $100, pays $10,000 with P = 0.0014 → E = 100(0.9986) − 9900(0.0014) = $86 per policy. **Normal:** bell, symmetric about μ, mean = median = mode, bends at μ ± σ, area 1; **Empirical rule** 68 / 95 / 99.7% within 1 / 2 / 3 σ. **Is it normal?** bell-shaped histogram or a straight normal probability plot → yes. Symbols: μ, σ population; x̄, s sample; σ² variance.